

AI BASED PLANT DISEASE CLASSIFIER

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Degree of Bachelor in Computer Science



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DEDICATION

To my loving family, whose unwavering support reminded me that courage and perseverance can turn dreams into reality.

DECLARATION OF AUTHORSHIP

We hereby affirm that the work presented in this thesis is entirely our own and has not been submitted for any degree or qualification at another university or institution.

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ABSTRACT

This project is on the development of an intelligent Plant Disease Classifier designed to support early detection and diagnosis of plant diseases in agricultural crops. The system aims to reduce crop losses and improve agricultural productivity by providing an automated, efficient, and user-friendly solution for identifying plant diseases using digital images of plant leaves. The proposed system is implemented as a web- and mobile-based application, enabling users to upload or capture leaf images and receive accurate disease classification results.

To achieve this objective, a machine learning–based classification model is employed. The system utilizes image preprocessing techniques such as resizing, normalization, and noise reduction to enhance image quality and improve model performance. A Convolutional Neural Network (CNN) algorithm is adopted due to its effectiveness in extracting spatial features from images. The CNN model is trained and validated using a labeled plant leaf image dataset, enabling it to learn disease-specific patterns and visual characteristics. Model training and evaluation are conducted using standard performance metrics, including accuracy and loss.

Experimental results and system testing demonstrate that the proposed model achieves reliable classification performance and effectively distinguishes between healthy and diseased plant leaves. The findings indicate that the Plant Disease Classifier provides a practical and scalable solution for automated disease detection in agriculture. Limitations of the system are discussed, and future improvements such as dataset expansion, real-time field application, and model optimization are recommended.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DFD	Data Flow Diagram
FAQ	Frequently Asked Questions
FAO	Food and Agriculture Organization
FPN	Feature Pyramid Network
IoT	Internet of Things
JSON	JavaScript Object Notation
KNN	k-Nearest Neighbors
LDA	Linear Discriminant Analysis
ML	Machine Learning
NMS	Non-Maximum Suppression
PCA	Principal Component Analysis
QA	Quality Assurance
R-CNN	Regions with Convolutional Neural Networks
REST	Representational State Transfer
RoIAlign	Region of Interest Align
SGD	Stochastic Gradient Descent

CHAPTER NO 1

INTRUDUCTION

1.1. Introduction

Agriculture is the backbone of many economies worldwide, providing food security, raw materials for industries, and employment for a significant portion of the population. However, the sustainability of agricultural production faces many challenges, one of the most critical being plant diseases. Plant diseases significantly reduce crop yield, deteriorate quality, and cause economic losses for farmers. According to the Food and Agriculture Organization (FAO), nearly 20–40% of global crop production is lost annually due to pests and diseases, making early and accurate detection a necessity rather than a choice. Among horticultural crops, apples, grapes, and cherries hold special significance due to their high economic value and widespread cultivation. Apples are one of the most consumed fruits globally and are cultivated in both temperate and subtropical regions. Grapes are not only consumed fresh but are also the raw material for juice, raisins, and wine production, making them highly profitable. Cherries, though cultivated on a smaller scale compared to apples and grapes, are valuable export fruits with high market demand. Despite their importance, all these crops are highly vulnerable to diseases such as apple scab, powdery mildew, grape black rot, grape downy mildew, and cherry powdery mildew, which primarily manifest on leaves before affecting fruits and stems.

Traditionally, diagnosing plant diseases has been performed by expert agronomists and pathologists who examine plant leaves visually and recommend treatment. While effective, this method is time-consuming, expensive, and inaccessible for many farmers in rural or resource-constrained areas. In addition, manual identification is prone to human error, as some diseases exhibit highly similar visual symptoms, such as yellowing, brown spots, or leaf curling. As a result, farmers often misdiagnose the disease, leading to improper treatment, excessive pesticide use, and further losses in yield. With the rapid advancement of artificial intelligence (AI) and computer vision, there has been increasing interest in developing automated systems to detect plant diseases from leaf images. Deep learning, particularly Convolutional Neural Networks (cnns), has shown remarkable performance in image classification tasks. Research studies have demonstrated that CNN-based models can achieve accuracy levels above 95% in distinguishing healthy and diseased plant leaves.

The availability of large, labeled datasets such as plantvillage has further accelerated this progress. Plantvillage is a public dataset consisting of thousands of leaf images captured under controlled conditions, covering multiple crops and their diseases. For this project, only color images of apple, grape, and cherry leaves were selected to ensure high-quality training data that reflects real-world farming conditions.

Despite these advancements, most existing research remains limited in scope. Many studies focus on a single crop, without addressing the need for a multi-crop solution that farmers with diverse cultivation require. Moreover, research often stops at developing models without moving toward real-world application deployment, leaving farmers with no accessible tools to benefit from these findings. Furthermore, language and usability barriers are rarely addressed. Farmers may not be familiar with scientific disease names or English-based descriptions, making these systems impractical for direct use. This project addresses these limitations by designing and developing a Plant Disease Classifier Application that focuses on apple, grape, and cherry leaves. The system integrates a trained deep learning model into a FastAPI backend and deploys it in a way that can be accessed through a user-friendly interface. The model classifies leaf images into healthy or diseased categories and outputs the result in two formats: scientific names in Latin, which provide accuracy and reliability for research and expert consultation, and treatment recommendations in the local language, which ensure that farmers clearly understand how to manage and prevent the spread of the disease.

The proposed application contains several important features that make it more practical and farmer-oriented compared to previous studies. It uses deep learning techniques, trained on plantvillage color images of apple, grape, and cherry leaves to achieve robust classification performance. To enhance the generalization capability of the model, preprocessing techniques such as resizing, normalization, and data augmentation are applied. Unlike single-crop approaches, this project includes three major fruit crops, offering a broader solution for farmers who often cultivate multiple fruits. The trained model is deployed using FastAPI, a high-performance Python web framework that allows real-time interaction through REST apis. This ensures that the

application is not just limited to research but can also function as a scalable service accessible on both web and mobile platforms. Another critical feature of the project is its multilingual output. The system not only provides the scientific Latin name of the detected disease but also delivers treatment recommendations in the local language, making it highly usable in rural communities. The treatment recommendation system goes beyond disease detection by offering actionable advice on fungicides, pesticides, organic remedies, and preventive measures, thus supporting farmers in making informed decisions.

Furthermore, the application has been designed with simplicity in mind. Its user-friendly interface allows farmers to capture or upload an image of a leaf using their phone's camera or gallery, after which the system processes the image, identifies the disease, and provides instant results. This immediate feedback mechanism empowers farmers to take timely action, reducing the risk of disease spread and minimizing crop losses. In addition to solving the current problem, the architecture of this system is future-proof and scalable. Although the current implementation focuses on apple, grape, and cherry leaves, it can easily be extended to other crops and integrated with additional features such as cloud storage and Internet of Things (iot) devices for real-time farm monitoring.

The motivation behind this project is to bridge the gap between research-level achievements and field-level usability. While many CNN-based models for plant disease classification exist in academic publications, very few are translated into real-world tools that farmers can actually use. By integrating scientific accuracy with practical usability, this project contributes to both computer science and agriculture. It ensures that farmers receive immediate and understandable results, while researchers and experts can also benefit from scientifically accurate disease labeling. This balance of scientific reliability and practical relevance sets the project apart from existing approaches. In summary, the introduction of this thesis emphasizes the importance of apples, grapes, and cherries in agriculture, highlights the devastating impact of plant diseases, reviews the limitations of traditional and existing AI-based detection methods, and presents the development of a Plant Disease Classifier Application as a

comprehensive solution. By combining deep learning, FastAPI deployment, multilingual outputs, treatment recommendations, and a user-friendly interface, this project not only contributes to academic knowledge but also addresses a real-world problem with direct social and economic impact.

1.2. Statement of the Problem

The production of apples, grapes, and cherries is a major source of income for farmers, yet these crops remain vulnerable to a range of leaf diseases that cause significant yield reduction and financial loss. Identifying these diseases early is often difficult because traditional diagnosis depends on the presence of agricultural experts, which is costly and not readily available in many farming regions. Manual inspection is also prone to error since many diseases share similar symptoms, making accurate detection even more challenging. Although advances in computer vision and deep learning have shown promise in classifying plant diseases with impressive accuracy, much of the existing work has been restricted to individual crops or limited to research settings without being translated into tools that farmers can practically use. This gap between technological potential and real-world application leaves growers without reliable and accessible solutions. Another challenge is that current systems rarely consider how results are communicated to farmers; while scientific terminology ensures precision, it may not be easily understood by those who most need the information. There is therefore a clear problem: farmers require a diagnostic approach that is not only accurate and efficient but also delivered in a form that they can use to make immediate decisions. This study seeks to address that problem by developing a multi-crop system tailored to apple, grape, and cherry diseases, offering both scientific reliability and practical accessibility, and in doing so, narrowing the divide between research innovation and everyday agricultural practice.

1.3. Significance of the Study

This study holds significance on multiple levels, both academic and practical. At the scholarly level, it contributes to ongoing work in artificial intelligence and agriculture by presenting a model that does not limit itself to one crop but instead demonstrates

how a unified approach can be applied across several fruit species. It further adds value by moving beyond experimental modeling to actual deployment, showing how image classification research can be translated into an operational system accessible to end users. On the practical side, the study provides farmers with an affordable and efficient way to detect diseases early, thereby reducing crop losses and improving overall productivity. For agricultural extension workers and policymakers, the system offers an additional resource that can strengthen training, disease management strategies, and decision-making in rural communities. The work also has broader societal relevance, as healthier crops mean improved food supply, greater income security for farming households, and reduced dependency on chemical treatments applied without proper diagnosis. Finally, for future researchers, the study establishes a framework that can be expanded to include other crops, languages, or integration with modern farming technologies such as cloud platforms and smart monitoring devices. By simultaneously advancing academic knowledge and offering a practical contribution to society, this study demonstrates its importance as both a research endeavor and a tool for positive agricultural impact.

1.4. Purpose of the Study

The main purpose of this study is to create an intelligent framework capable of identifying diseases in apple, grape, and cherry leaves with high reliability by applying modern deep learning methods. Rather than remaining as a purely experimental model, the project is designed to evolve into a working application that can serve both academic research and agricultural practice. By transforming image data into meaningful diagnostic results, the study aims to offer a tool that farmers and agricultural stakeholders can access easily through a lightweight digital platform. Another purpose is to bridge the gap between the scientific community and farming communities by ensuring that the system communicates results in a way that is both technically accurate and practically useful. In doing so, the research seeks not only to demonstrate the potential of AI in agriculture but also to provide a foundation that can be expanded in the future to other crops and regions.

1.5. Research Questions

This study seeks to answer the following questions: How can a convolutional neural network–based deep learning model be designed and trained to accurately classify diseases in apple, grape, and cherry leaves using leaf image data? How can the trained deep learning model be effectively integrated into a web- and mobile-based application to provide real-time plant disease diagnosis? How can disease classification results be presented in a way that maintains scientific accuracy while remaining understandable and usable for farmers through localized language and practical treatment recommendations? What potential benefits can such an AI-based plant disease classification system offer in supporting early disease detection and improving agricultural decision-making?

1.6. Expected Outcome

The expected outcome of this work is a robust and practical system that demonstrates how artificial intelligence can contribute to crop protection and sustainable farming. The model is anticipated to achieve strong performance in detecting leaf diseases across multiple fruit crops, showing that a multi-crop approach is both feasible and effective. Alongside technical accuracy, the study expects to produce a functional application that provides users with timely feedback in a format they can understand and act upon. This outcome should contribute to reducing misdiagnosis, improving early intervention, and ultimately minimizing crop losses. From a scholarly perspective, the research is expected to add to existing literature by illustrating how computer vision models can be transitioned into practical tools with real-world impact. From a societal perspective, it is expected to empower farmers, support agricultural extension programs, and encourage future innovation by demonstrating a scalable solution that others can refine and expand.

1.7. Summary

Plant diseases represent a significant threat to agricultural productivity, particularly for high-value horticultural crops such as apples, grapes, and cherries, which contribute

substantially to farmers' income and global food supply. Traditional methods of disease diagnosis rely on expert visual inspection, a process that is often costly, time-consuming, and inaccessible in many agricultural regions. Moreover, visual similarity among disease symptoms increases the likelihood of misdiagnosis, leading to ineffective treatment and increased crop losses. Although recent advances in artificial intelligence and deep learning have demonstrated strong potential for automated plant disease detection, existing studies frequently remain limited to single-crop scenarios, experimental settings, and lack practical deployment and usability considerations for farmers.

This study addresses these limitations by developing a multi-crop Plant Disease Classifier Application that utilizes deep learning techniques to identify diseases in apple, grape, and cherry leaves. The model is trained on color images from the PlantVillage dataset and deployed through a FastAPI-based backend to enable real-time interaction. The system is designed to provide scientifically accurate disease identification alongside treatment recommendations presented in the local language, thereby balancing technical precision with practical accessibility. A user-friendly interface allows farmers to capture or upload leaf images and receive immediate diagnostic feedback.

CHAPTER NO 2
LITERATURE REVIEW

2.1. Introduction

This chapter presents a comprehensive review of existing research, systems, and practices related to plant disease detection, with a particular focus on the role of technology- especially smartphones- in facilitating these processes. The purpose of this review is to establish a solid foundation for understanding the current state of work in this field and to identify the limitations and gaps in existing systems that our proposed solution aims to address. By examining previous studies and technologies relevant to plant disease detection applications, this chapter not only highlights the progress made so far but also provides the conceptual and technical framework that supports the development of our proposed system.

2.2. Literature Review

The author of [1] conducts a comprehensive survey of deep learning methods for plant disease detection. The authors aggregate studies utilizing CNN architectures (including AlexNet, VGG, ResNet, and MobileNet). This review provides readers with a thorough overview of how deep learning is applied in plant disease identification.

The review employs an analytical and comparative approach. It contrasts various studies' outcomes and shows that CNNs yield significantly higher accuracy than previous techniques. The paper also details the datasets, training methods, and evaluation metrics used.

In [2], a machine learning technique for plant disease detection is developed, aiming to provide a cost-effective, automated diagnostic tool for farmers. The authors utilize a Random Forest classifier. The workflow consists of four stages: image preprocessing, feature extraction (using Histogram of Oriented Gradients, Hu moments, Haralick texture features, and color histograms) to form feature vectors, followed by model training.

Images of 160 papaya leaves, both healthy and diseased, were used for training. Compared to other methods such as SVM and Logistic Regression, Random Forest performed the best, achieving an accuracy of nearly 70%.

Overall, the article demonstrates that classic machine learning methods, such as Random Forest, offer a basic approach; however, their accuracy is lower than that of deep learning.

The performance of machine learning and deep learning methods to determine which approach is better at identifying plant diseases is compared [3]. They used the PlantVillage dataset and trained both classical machine learning methods (SVM, Random Forest, kNN) and CNN architectures on it. According to the results, CNN achieved an accuracy above 95%, while machine learning models did not reach this level.

In contrast with others in [4], they have compared the performance of various deep learning models, such as AlexNet, VGG, ResNet, Inception, and DenseNet, in plant disease detection.

They used PlantVillage and crop image data, pre-processed it, and then trained the models. The results showed that ResNet and DenseNet achieved the highest accuracy, and transfer learning was more effective for small data sets.

Team [5] studies how image recognition models can help identify leaf diseases in crops with deep learning. They pointed out that plant diseases can harm both crop yields and food quality. Since traditional methods are slow and need experts, they tested pre-trained CNN models like VGG-16, ResNet-50, Inception V4, and DenseNet-121 to see which worked best.

They used the PlantVillage dataset, which contains over 54,000 leaf images (38 categories of diseased and healthy leaves). They processed the images (resizing, normalizing, and data augmentation) and then tuned the hyperparameters of the models during training. The results showed that all models had high accuracy, but DenseNet-121 performed the best, achieving almost perfect accuracy (99.81%).

The study to compare advanced deep learning models for plant disease detection was conducted in [6] that their goal was to find a system that would be fast, accurate, and useful in agriculture. They used popular CNN models such as VGG16, Inception V4, ResNet (50, 101, 152), and DenseNet-121. All models were fine-tuned with pre-trained weights from ImageNet and tested on the PlantVillage dataset.

They preprocessed the dataset (resized and normalized) and divided it into training, validation, and testing parts. During training, they modified the last layers and calibrated them using SGD. The results showed that all models had high accuracy, but DenseNet-121 performed the best (99.75% accuracy). This model required less computational power than the others and did not suffer from overfitting. In contrast, VGG16 was the weakest.

As stated in [7] they investigated CNN-based plant disease detection. Their goal was to show how accurately CNN could detect plant diseases.

They used the PlantVillage dataset, which contains over 50,000 images. They tested the AlexNet and GoogLeNet models and achieved an accuracy of up to 99.35%.

Explain a method for identifying plant diseases based on deep learning, as pointed out in [8]. The researchers used the PlantVillage dataset, which contains thousands of images of healthy and diseased leaves. They performed pre-processing and augmentation of the images and then trained a disease detection model using convolutional neural networks (CNNs).

The results showed that the CNN models achieved an accuracy of more than 98% in identifying plant diseases. In line with findings from [9], the process of diagnosing agricultural plant diseases is improved using convolutional neural networks (CNNs). Traditional diagnostic methods are labor-intensive, expert-dependent, and impractical for large farms. To address this, CNNs are trained using healthy and diseased leaf images from the PlantVillage dataset. Preprocessed images were augmented to enhance resilience, and models were trained on several crops and diseases. Findings were higher achievements in classification accuracy as compared to the conventional diagnostic approaches. This indicates that CNNs can be dependable instruments with regard to precision agriculture.

According to [10], they tackle the subject of sustainable agriculture in the context of climate change and strive to enhance crop monitoring algorithms by using machine learning and remote sensing. Using several accumulating spectral and hyperspectral satellite data images/UAV data, and combining with ML algorithms, they identified the indicators of crop growth.

It was demonstrated that the trained models strongly performed at aiming stress on plants, crop type classification, and yield prediction, offering high accuracy and scalability compared with traditional methods of analysis.

The main drawbacks are that these methods need large, high-quality datasets, which are often hard to get in developing regions. They are also sensitive to the environment, can have sensor inconsistencies, and require expensive computing. Most tests were done in controlled settings. Future work will focus on using bigger datasets, adding more field measurements, and creating cheaper solutions to make the technology more accessible.

Selected a specific crop, apple trees, and studied them in particular as they are susceptible to the leaf disease [11]. They observed that the conventional means of detecting those problems were slow and at times inaccurate, particularly in situations when the leaves contained minute spots or were photographed in chaotic, real-world conditions. As a solution, they scaled up the Faster R-CNN model and used it with Res2Net and a Feature Pyramid Network (FPN) to make the system capable of locating finer details in leaf images.

Unlike other works, what is outstanding about their work is that they did not use clean lab images. Rather, they made up their own data, AALDD, stuffed with thousands of actual photographs of the leaves of apples in orchards. They also implemented methods such as the RoIAlign that helped to sharpen the capability of the model in feature location and soft-NMS to minimize errors when disease spots were overlapping. Results were encouraging: for diseases such as powdery mildew and mosaic, this model reached more than 90 percent accuracy, and it demonstrated that it can also deal with real farming conditions. Nevertheless, it had certain challenges. This model did not cope with rust disease, which appears as extremely

small and fragile spots, and its speed (approximately 12 frames per second), which is too slow for real-time use in the field. While this is a limitation, the research still marks important progress. It shows that deep learning can make disease detection easier for farmers if future models become faster and more efficient.

A different category of devotees shifted their focus to cherry production in Chile, an important high-value crop where precise predictions of harvest are worth billions of dollars. Their designed system was based on the Faster R-CNN model with the help of Inception V2 and made to recognize cherries on field images at varying lighting conditions and maturity stages [12].

The error rate of the forecasting was halved, the success rate of the older techniques being at only 25 percent of the forecasting error rate previously. Farmers themselves could now make very reliable forecasts of the amount of yield they would get, which helped them to organize labor, storage, and market plans more predictively. Naturally, there were hindrances. The model was more difficult to recognize smaller cherries, and the model would occasionally lead to errors, whereby leaves or other objects made contact with the images. Nevertheless, the findings reveal that deep learning would not only transform agricultural forecasting but also the forecasting of most crops, in addition to cherries. The paper identifies the ways AI can add real wealth to the agricultural sector through more accurate predictions and less uncertainty.

Similarly, [13] discovered bacterial diseases through an analysis of the images of the leaves of cherries. They did not create a model; instead, they applied DarkNet-19, a pre-trained deep learning model, to extract features. These characteristics were then injected into various machine learning methods, which were the Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), and K-Nearest Neighbor (KNN).

The sample consisted of 1,906 images, and it included healthy, mildly diseased, and severely diseased. These findings were promising: the system was capable of identifying bacterial disease at about 88 percent, although the highest accuracy was achieved by SVM. This stood out because it was not only possible to draw the line

between a healthy condition and a sick one, but also to define the various levels of disease severity, which could be important in a quick response to it.

This perspective is further reinforced in [14], the identification of cherry bacterial leaf diseases. The focal point, this time, was the application of deep features made by DarkNet-19 together with conventional machine learning classifiers like SVM, KNN, and LDA. The dataset was identical in size to that used previously- 1,906 leaf images categorized as healthy, mildly diseased, and severely diseased. The strategy was actually highly successful, as it was able to detect disease severities with an accuracy of approximately 88 percent, and SVM outperformed it once again. This lesson was that with the standard ML classifiers and deep learning to extract features, the system was capable of providing reliable disease detection without an overly complex architecture.

The authors recognized that the outcome would be given a higher quality of preprocessing and tuning, but on their side, they also indicated the extent to which such a system could be of significance in practice. The timely detection of the disease gives the farmers an earlier opportunity to do something and save their yields and enhance their productivity. They indicated that in the future, they might increase the dissemination of the system to a larger number of plant species, disease types, and possibly even modules that could interact with the treatment plan and diagnosis.

As noted in [15], they develop four deep learning models: a custom “Vanilla” CNN, and three other models that improved via transfer learning from pre-trained architectures (VGG16, MobileNet, and AlexNet). They created a data set of grape leaf images, which was pre-processed through (cropping, flipping), to balance the classes. The models were trained and evaluated on the dataset. Finally, they created an ensemble model that predicted all four models by using an average voting method to boost accuracy.

The meteorological data of (temperature, humidity, rainfall) and the historical disease occurrence records are collected [16]. They used MATLAB for data preprocessing, which included outlier detection and visualization with Principal Component Analysis.

They have built and compared several prediction models. The best-performing model was a pruned decision tree that used rules based on weather parameters and the month of the year to predict disease risk. This paper predicts the risk of two cherry fruit diseases by using weather data and data mining techniques, rather than image analysis.

The challenge of the class imbalance in the plant disease dataset is addressed by proposing three novel methods based on the DenseNet-121 architecture to improve the recognition of six types of apple leaf diseases. In [17], they used standard multi-class classification with a cross-entropy loss function, and they proposed three alternative approaches: (Regression, Multi-label Classification, and Focal Loss).

As revealed in [18], they aim to develop a highly accurate yet computationally efficient model for grape leaf disease classification that can be deployed on mobile devices with limited resources.

The authors chose the lightweight CNN architectures (ShuffleNet V1 and V2) as their backbone. To boost accuracy without a significant computational cost, they integrated a Channel-wise Attention (CA) mechanism into these networks. This one allows the models to focus on the information features in the image. They tested their improved models on a grape leaf dataset and compared their performance and size speed against other models like AlexNet and MobileNet.

As proposed in [19], an automated system for classifying disease in grape leaves using an ensemble of deep learning models. The goal of that is to improve accuracy and reduce overfitting compared to using individual models.

And they use an ensemble learning approach that combines three pre-trained convolutional Neural Networks (CNNs): VGG16, VGG19, and Xception. And the data set they used was the publicly available PlantVillage dataset, specifically the 4062 images of grape leaves.

A real-time object detection system to identify and locate multiple diseases on a single apple leaf image in complex field conditions was developed in [20]. The focus on real-time performance and handling small disease spots against complex backgrounds.

They construct their own Apple Leaf Disease Dataset (ALDD) with 26,377 images from both laboratory and real-field conditions, which are manually annotated. They create a new Model Architecture that is INAR-SSD, which is trained on their own dataset, that is the ALDD dataset.

As explored in [21], creating a highly accurate yet lightweight and efficient model for classifying grape diseases makes it suitable for deployment on devices with limited computational resources. The proposed model (GLD-DTL) with a very small model size and a significantly larger output performed is larger than models like VGG19. They used the model of MobileNetV3, a model designed for mobile devices, as their base. They use data augmentation to balance the classes. The goal is to use a model that is both small in size and highly accurate. The model's performance is demonstrated on a cleaned and curated dataset. Its robustness to very noisy, low-quality field images is implied but not extensively tested.

As examined in [22], they implement a custom Convolution Neural Network (CNN) from scratch to classify four types of apple Leaf conditions, focusing on a simple yet effective approach.

The aim is to build a functional model without relying on pre-trained networks or complex architectures. They use a subset of the PlantVillage dataset containing 2,526 images. The model is trained from scratch using the Adam optimizer. And it identifies the dataset's artificial nature as the biggest drawback for real-world applications. Parameters like dropout, batch size, and train-test split ratio.

They develop a highly accurate and efficient image classification model for detecting grape leaf diseases using the lightweight MobileNetV2 architecture, aiming for potential deployment on resource-constrained devices in [23]. The primary goal is to demonstrate that a lightweight CNN like MobileNetV2 can achieve superior accuracy while maintaining computational efficiency. They employ transfer learning with the MobileNetV2 model pre-trained on ImageNet. The top layers are replaced with custom layers for the 4-class problem. The external validation set was very small

(20 images); a larger and more diverse external dataset is needed for a robust assessment of generalization.

As demonstrated in [24], they addressed the complex problem of detecting and localizing multiple diseases on a single leaf image, using an object detection approach, rather than standard image classification, which assumes one disease per image. The R-CNN models are trained to both classify the disease within each proposed region and predict the precise bounding box coordinates. The research aims to detect and draw bounding boxes around three grape leaf diseases (Black Rot, Black Measles, Isariopsis) and healthy leaves, even when they appear together in a single image. A key feature is the inclusion of a confidence value for each prediction. The R-CNN models are trained to both classify the disease within each proposed region and predict the precise bounding box coordinates. They compare three pre-trained networks (AlexNet, GoogLeNet, ResNet-18) as the backbone feature extractors within a Regions with Convolutional Neural Networks (R-CNN) framework for object detection.

2.3. Summary

The literature review of the thesis on plant disease detection Application provides an overview of the existing literature on plant disease detection systems and Android-based mobile apps. This literature review examines 5 sources, including research papers, videos, and websites, to explore the development and impact of plant disease detection applications. various studies and journals were analyzed to address the significance of this issue.

Most existing studies on plant disease detection have demonstrated that deep learning, particularly CNN-based models such as ResNet, DenseNet, and MobileNet, can achieve very high accuracy using datasets like PlantVillage. However, these works are mostly limited to lab-controlled images, require large labeled datasets, and rarely focus on deployment for farmers. Unlike these studies, our project addresses these gaps by developing a lightweight mobile application for three key fruits (apples, cherries, and grapes) that not only detects diseases but also provides localized treatment recommendations in Pashto/Dari, making it more practical and accessible for farmers in Afghanistan.

CHAPTER NO 3

METHODOLOGY

3.1. Introduction

This chapter focuses on the research methodology adopted for the development of the Plant Disease Classifier system. The methodology outlines the step-by-step process that guided the research from data collection to model evaluation and system implementation. It provides a comprehensive explanation of how the project objectives were achieved through systematic planning, analysis, design, and testing. The methodology is crucial because it ensures the research is conducted in a logical, reliable, and reproducible manner.

3.2. Research Method

This study employs a quantitative research design based on a Secondary Data Analysis approach. The primary objective of this phase was to acquire a large, diverse, and reliably labeled dataset of images to facilitate the development and training of a robust Convolutional Neural Network (CNN) model. This method was selected over primary data collection (such as taking our own photographs) to ensure access to a wide variety of disease manifestations, lighting conditions, and fruit growth stages, which is crucial for building a generalizable AI model.

3.2.1. Data Source and Collection Procedure

The data for this project was sourced from pre-existing, publicly available image repositories and benchmark datasets commonly used in computer vision and agricultural informatics. Key data sources included:

The PlantVillage Dataset: A widely recognized benchmark in plant disease detection, providing thousands of labeled images of healthy and diseased crops. Kaggle Repositories: Specific datasets containing annotated images for apple, cherry, and grape diseases were sourced from this platform. Academic Research Databases: Images from published papers in domains of precision agriculture and plant pathology were utilized where permissions were granted.

3.2.2. Data Description and Selection Criteria

The collected archival data consists of digital images of apple, cherry, and grape leaves and fruits. Each image is annotated with labels indicating the plant species and its health status.

Target Classes: The model was designed to distinguish between the following states for each fruit type: Apple: Healthy, Apple Scab, Black Rot; Cherry: Healthy, Powdery Mildew, and Grape: Healthy, Black Rot, Esca (Black Measles), Leaf Blight.

Inclusion Criteria: Images were selected based on clarity, confirmed diagnosis, and relevance to common diseases affecting the target fruits. **Data Volume:** A total of 11,336 images were compiled and formed the core dataset for model development.

3.2.3. Ethical Considerations in Secondary Data Use

All data used in this study were obtained from public sources that grant explicit permission for use in academic and non-commercial research. We ensure compliance with the original licenses of the datasets, and proper attribution will be provided in the final research documentation.

3.3. Software Development Model

In our project, we used the Agile Software Development Model. This model was chosen because it allows flexibility, continuous improvement, and collaboration throughout the development process. The Agile approach helped our team to deliver the project in multiple small Sprints instead of completing it all at once.

Sprint 1: Project Setup & Planning

Table 3.1 Sprint 1: Project Setup & Planning

User Story	Key Tasks / Activities	Story Points	Priority	Owner
Project Initialization	Define system architecture, modules (ML, API, UI), and Agile workflow	3	High	PO / Dev Lead
Repository Setup	Create GitHub repo, branch structure, and commit policy	1	High	DevOps
Environment Configuration	Set up Python, TensorFlow, FastAPI, and supporting libraries	3	High	DevOps / ML Engineer
Dataset Identification	Select reliable secondary dataset PlantVillage	2	High	Data Engineer
Documentation Setup	Prepare initial documentation and sprint backlog	2	Medium	PO / Scrum Master

Sprint 2: Dataset Preparation & Preprocessing

Table 3.2 Sprint 2: Dataset Preparation & Preprocessing

User Story	Key Tasks / Activities	Story Points	Priority	Owner
Dataset Acquisition	Download and verify secondary dataset	2	High	Data Engineer
Data Cleaning	Remove noisy or duplicate images	3	High	Data Engineer
Data Labeling & Organization	Ensure all images are properly labeled per disease type	2	High	Data Engineer
Data Augmentation	Apply transformations (rotation, zoom, flip) to balance dataset	5	High	ML Engineer

Dataset Documentation	Record dataset source, license, and preprocessing details	2	Medium	Data Engineer
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Sprint 3: Model Development & Training

Table 3.3 Sprint 3: Model Development & Training

User Story	Key Tasks / Activities	Story Points	Priority	Owner
Model Architecture Design	Choose and define CNN architecture MobileNetV2	4	High	ML Engineer
Model Training	Train CNN on preprocessed dataset	8	High	ML Engineer
Model Evaluation	Evaluate accuracy, precision, recall, F1-score	3	High	ML Engineer
Model Optimization	Perform hyperparameter tuning and prevent overfitting	4	Medium	ML Engineer
Model Saving	Save best model .keras for integration	2	Medium	ML Engineer

Sprint 4: Integration & Application Development

Table 3.4 Sprint 4: Integration & Application Development

User Story	Key Tasks / Activities	Story Points	Priority	Owner
Backend Development	Build FastAPI for model prediction	4	High	Backend Dev

Frontend Development	Build simple UI for image upload and displaying disease results	5	High	Frontend Dev
API Integration	Connect frontend with backend API for prediction	3	High	Full Stack Dev
CI/CD Pipeline	Automate build and deployment	3	Medium	DevOps

Sprint 5: Testing, Deployment & Maintenance

Table 3.5 Sprint 5: Testing, Deployment & Maintenance

User Story	Key Tasks / Activities	Story Points	Priority	Owner
Testing & QA	Perform unit, integration, and user acceptance testing	4	High	QA Engineer
Deployment	Deploy system using Render	4	High	DevOps
Documentation	Prepare technical, dataset, and user manuals	3	Medium	PO / Dev Lead
User Feedback	Collect and analyze user feedback for improvement	2	Medium	All Team Members
Maintenance & Updates	Plan for bug fixes, model retraining, and future upgrades	3	Medium	DevOps / PO
Sprint Review & Retrospective	Review achievements and next iteration improvements	1	Medium	Scrum Master

3.4. System Design Model

The System Design Model defines how the system components interact and how the workflow of the plant disease classifier is organized. It provides a detailed structural and behavioral understanding of the system.

3.4.1. DFD Level 0:

This diagram shows the overall system boundary and external interactions. It depicts the Farmer as the primary user interacting with the Plant Disease Classifier System, which exchanges data with the Static JSON File containing disease treatment information. The diagram illustrates high-level data flows without detailing internal processes.

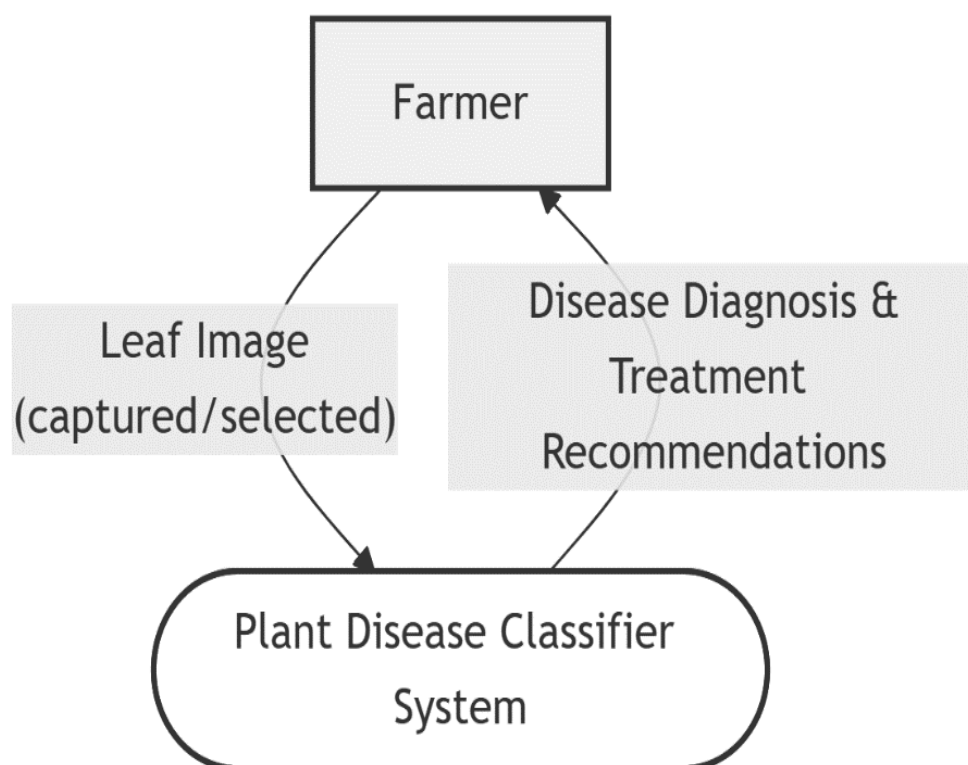


Figure 3.1 DFD Level 0

3.4.2. DFD Level 1:

This diagram breaks down the main system into sub-processes showing how data moves through the system. It includes processes like Image Capture & Selection, Mobile Image Preprocessing, Image Reception & Server Processing, Disease Prediction, Treatment Data Retrieval, and Results Display. It demonstrates the sequential flow of data from image input to treatment recommendations.

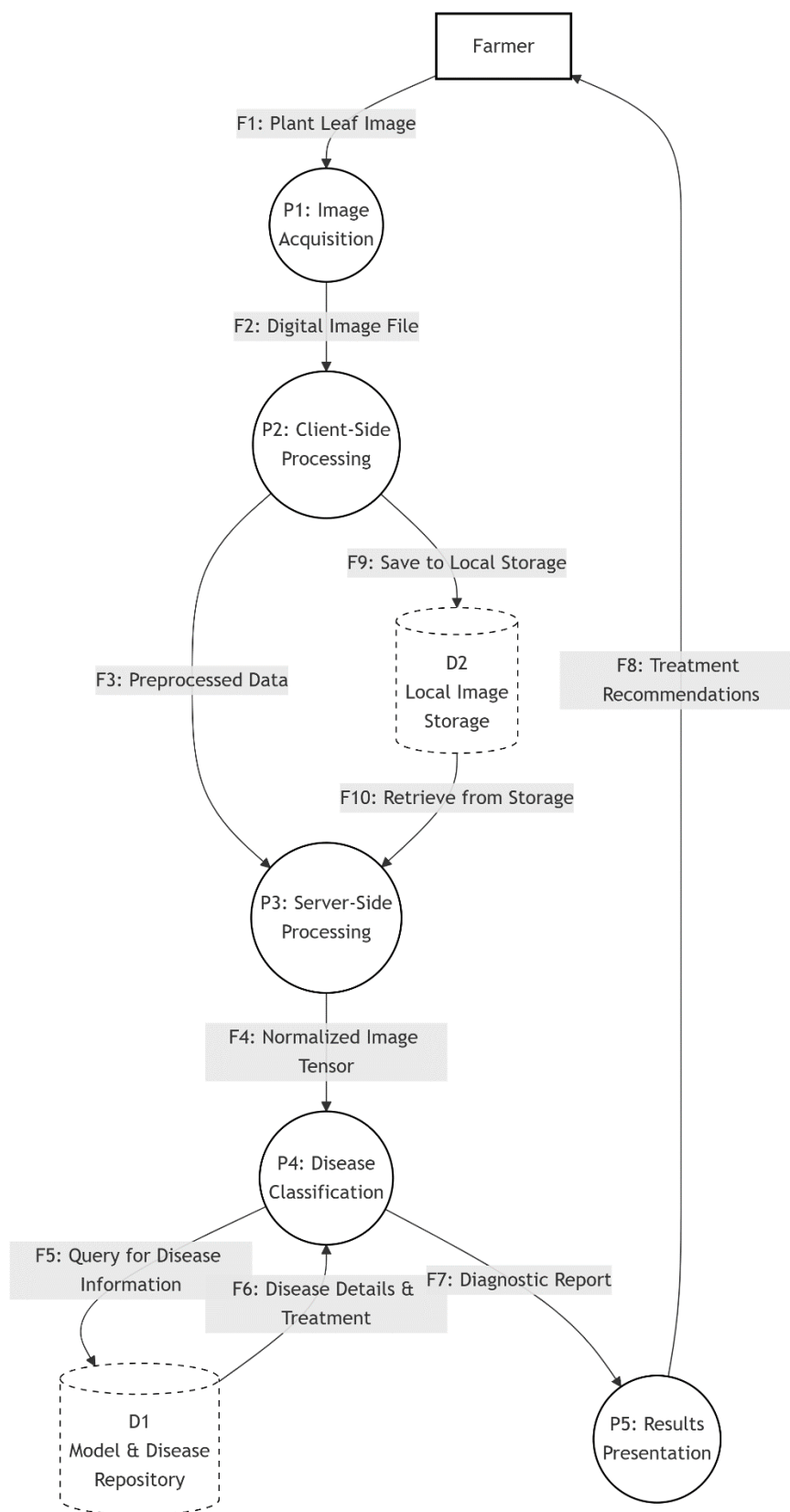


Figure 3.2 DFD Level 1

3.5. Use case Diagram:

This diagram illustrates all the functionalities available to the Farmer user. It shows use cases such as Capture Leaf Image, Upload Image, Preprocess Image, Classify Disease, Retrieve Treatment Info, and View Diagnosis Result. The diagram uses standard UML notation with include relationships showing mandatory connections between different functionalities.

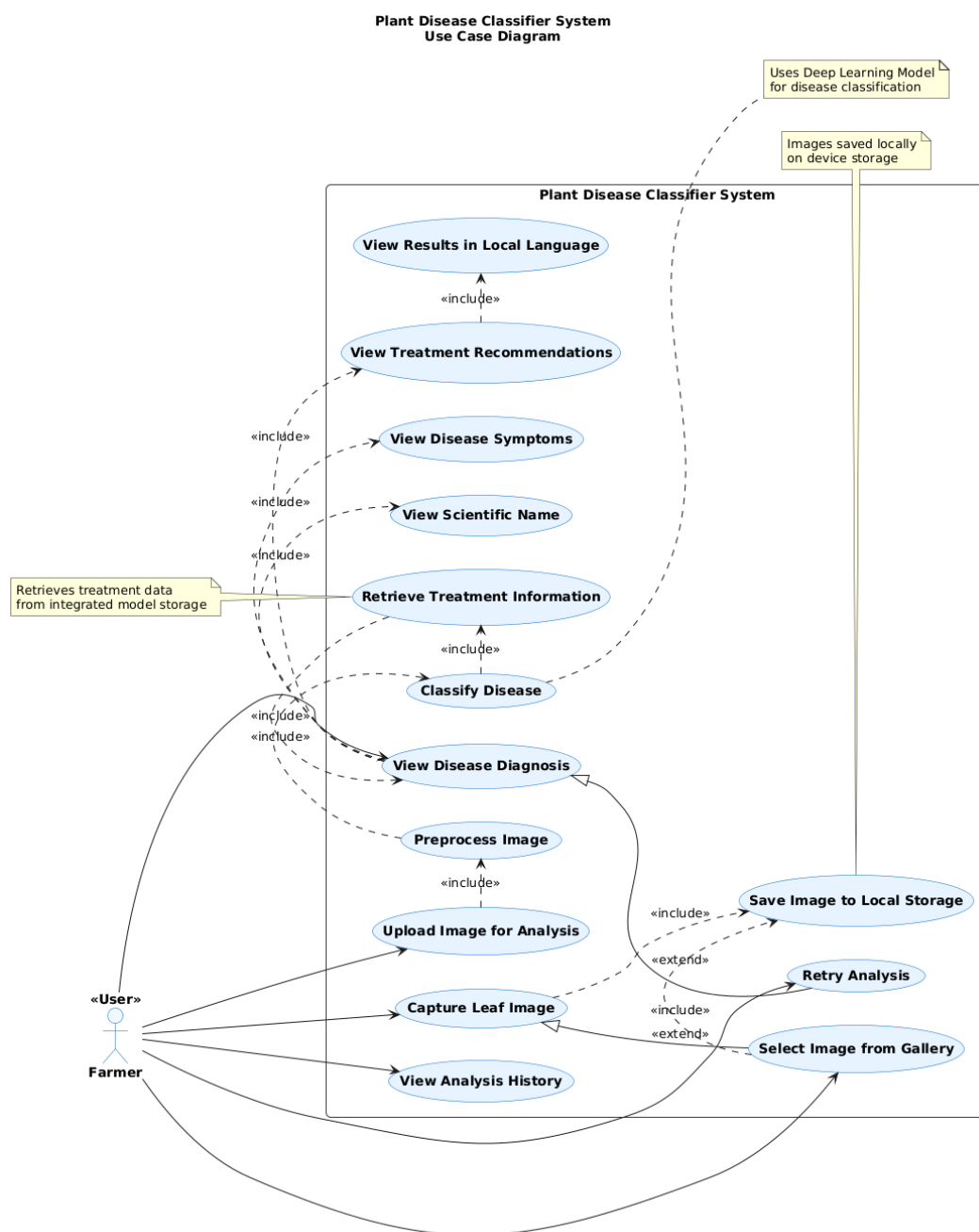


Figure 3.3 Use case Diagram

3.6. Time Sequence Diagram:

This diagram displays the chronological sequence of interactions between system components over time. It shows the step-by-step process starting from the Farmer capturing an image, through mobile app processing, backend analysis, disease classification, treatment retrieval, and finally displaying results back to the user. The vertical timeline and horizontal participants clearly show the order of operations and communication patterns.

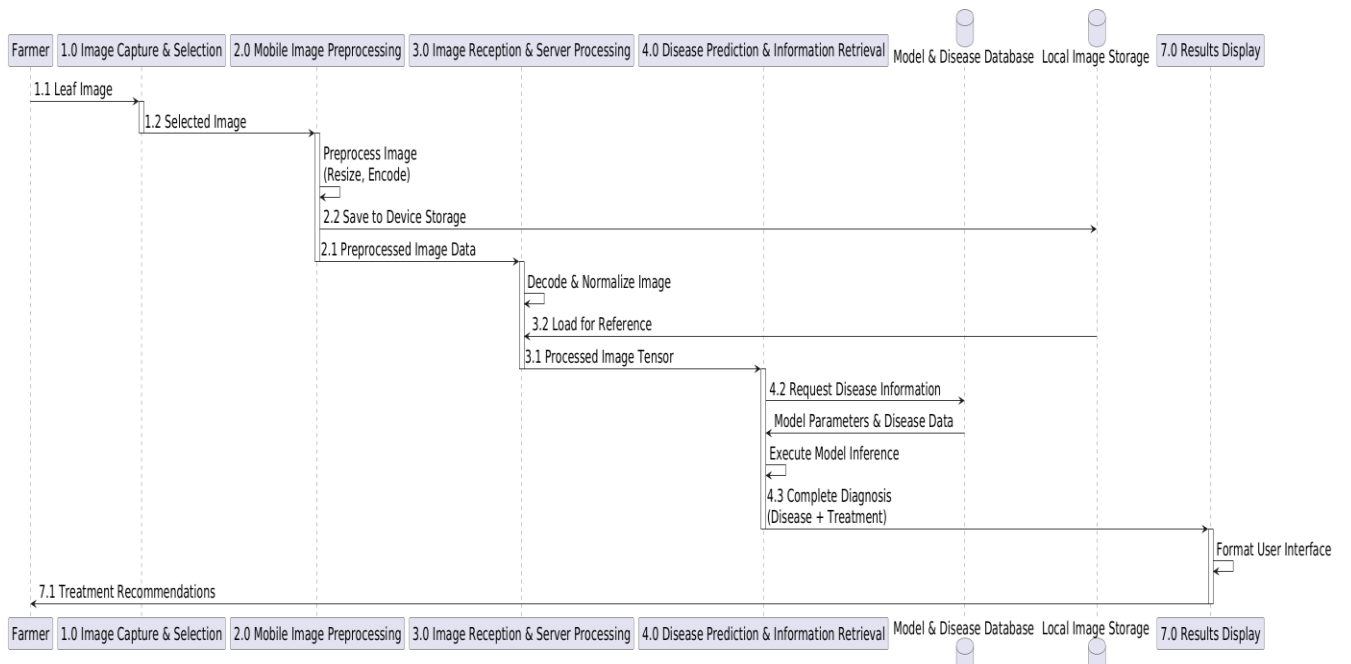


Figure 3.4 Time Sequence Diagram

3.7. Data Analysis

The analysis focused on model training and evaluation. A Convolutional Neural Network (CNN) architecture was constructed and trained on the prepared training set, allowing the model to learn discriminative features for classifying the health status of apples, cherries, and grapes. The model's performance was rigorously assessed using the held-out test set. Key quantitative metrics, such as Accuracy, Precision, Recall, and F1-Score, were calculated to quantitatively measure the model's classification performance. Furthermore, a confusion matrix was generated to provide detailed

insight into the model's strengths and weaknesses across different fruit types and disease classes

CNN algorithm Performance Summary

Accuracy: 89 Precision: 91 Recall: 89 F1-Score: 89

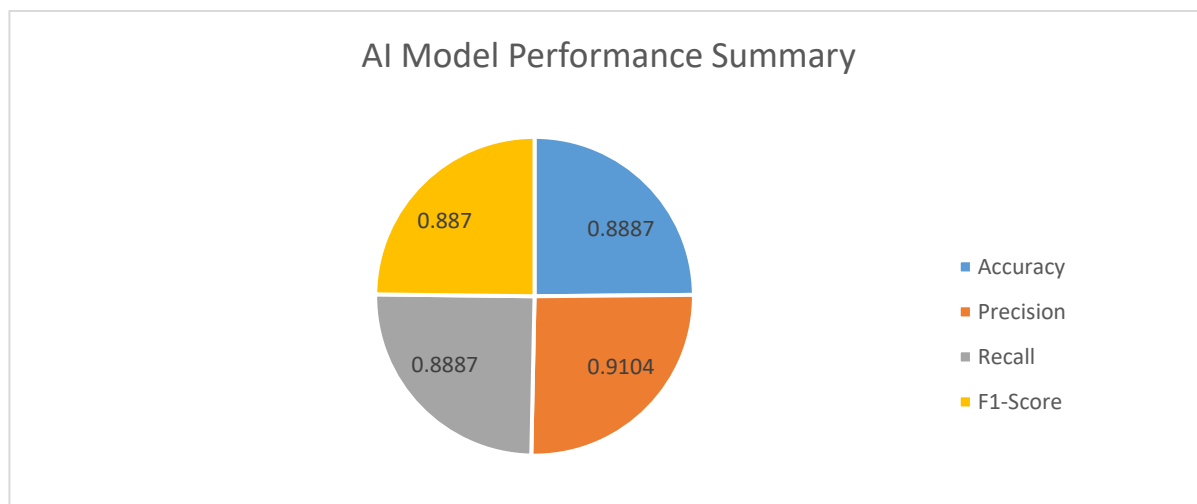


Figure 3.5 Data Analysis

3.8. Tools Used

The plant disease classification system leverages a robust technology stack engineered for machine learning and image processing tasks. Python serves as the core programming language, providing extensive libraries and frameworks for AI development. The deep learning components are built using TensorFlow with Keras for constructing and training convolutional neural networks on plant imagery. Pillow handles critical image preprocessing operations including resizing, normalization, and augmentation of leaf images. For data manipulation and numerical computations, NumPy and Pandas facilitate efficient dataset management and feature engineering. Model performance evaluation and data visualization are accomplished through Matplotlib, enabling clear interpretation of training metrics and results. The backend API is developed with FastAPI to create a high-performance web interface that handles image uploads and prediction requests. Mobile application development is supported by Android Studio for creating the farmer-facing interface, while VS Code serves as the primary integrated development

environment for coding, debugging, and version control throughout the project lifecycle. This comprehensive toolset ensures seamless integration between components while maintaining scalability and performance across the entire system architecture.

3.9. Summary

This chapter described the system design and implementation of the Plant Disease Classifier. It detailed the working of the system using Use Case, DFD, and Sequence diagrams, discussed how data analysis was performed, and listed the tools used to build the model. The system's architecture ensures efficient data flow and accurate disease classification. By combining deep learning techniques with image processing, the model provides a reliable way to detect plant diseases and help farmers maintain healthy crops

CHAPTER NO 4
PROPOSED SOLUTION

4.1. Introduction

This chapter presents the implementation of the Plant Disease Classifier system, which includes both the mobile application developed using Android Studio and the web application created with HTML, CSS, and JavaScript. The purpose of this chapter is to demonstrate how the designed system was transformed into a working software solution, explain its core functionalities, and show the results obtained during testing.

The mobile application enables users—such as farmers, students, and agricultural experts—to capture or upload leaf images directly from their Android devices. These images are processed by the trained machine learning model to identify plant diseases quickly and accurately. The interface is designed to be simple, responsive, and user-friendly.

In addition to the mobile app, a web-based interface has been developed to provide users with an alternative platform for accessing the classifier. The web application allows image upload through a browser and displays disease predictions in real time. This enhances accessibility by supporting users who prefer desktop or laptop usage.

4.2. Proposed System

The proposed system is an intelligent Plant Disease Classifier that automatically detects plant diseases from leaf images using a trained machine learning model. It provides fast, accurate, and accessible disease diagnosis through two platforms: a mobile app and a web app. The mobile application, built in Android Studio, allows users to capture or upload leaf images and instantly receive disease predictions, making it suitable for real-time field use. The web application, developed using HTML, CSS, and JavaScript, enables users to upload images through a browser and get quick predictions, offering convenience for desktop and laptop users.

4.2.1 Home Screen Page

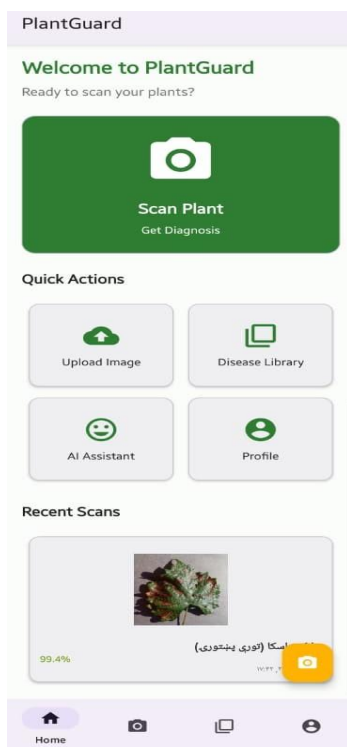


Figure 4.1 Home Screen Page

These screens provide the main features of the app. The user can choose to upload an image, capture a photo, or browse previous predictions. The design is simple to ensure easy navigation.

4.2.1. Image Upload / Capture Screen

These screens allow the user to select an image from the gallery or take a photo using the camera. Selected images are previewed before prediction. The system ensures the user chooses a clear leaf image for accurate classification

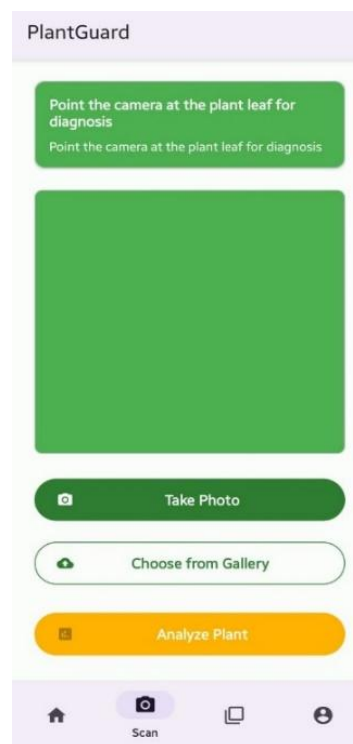


Figure 4.2 Image Upload / Capture Screen

4.2.2. Result Page

The page is result page serves to clearly present the model's diagnostic conclusion and facilitate user action. It displays the detected condition with a corresponding confidence score, organizes detailed supporting information into structured sections for clarity, and provides direct controls for the user to either initiate a new scan or save the current analysis report for future reference.



Figure 4.3 Result Page

4.2.3. Disease Library

This page is Disease Library allows to search and browse detailed information on various plant diseases and conditions. Identify problems, learn about causes and symptoms, and discover the right treatments to keep your plants thriving.



Figure 4.4 Disease Library

4.2.4. Disease Details Screen



Figure 4.5 Disease Details Screen

This section shows detailed information about the identified disease, including causes, symptoms, environmental factors, and recommended solutions. This helps users understand the disease more deeply.

4.2.5. My Profile Page

The Profile section is the personal command center and digital greenhouse within the Plant Guard app. It serves as a customized space where users manage their identity, their plant collection, and their overall interaction with the application.

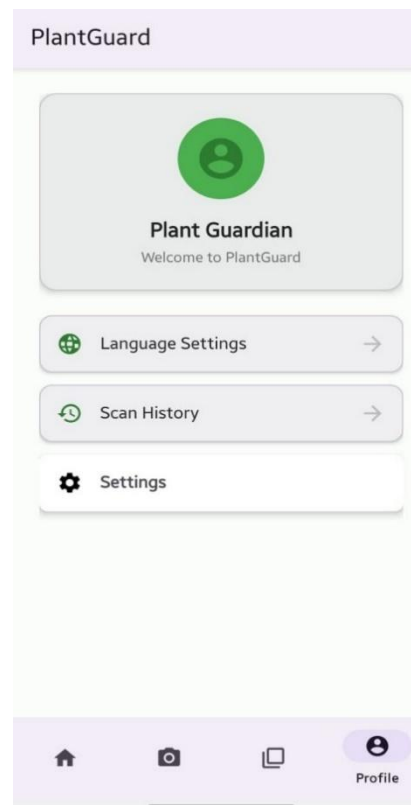


Figure 4.6 My Profile Page

4.2.6. App Setting

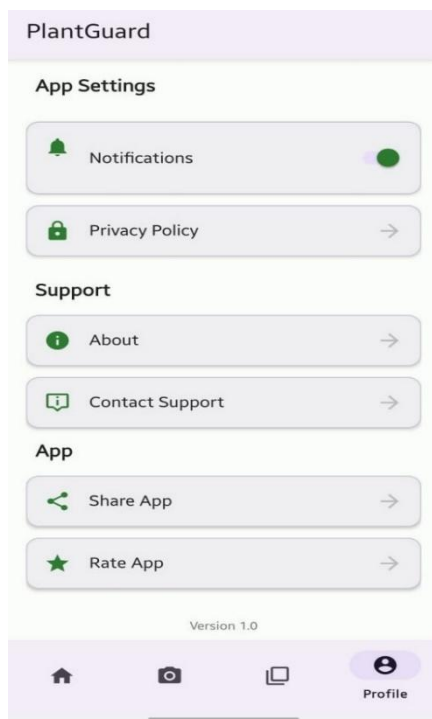


Figure 4.7 App Setting

In App Setting section in my app's control center. Here, I can personalize notifications, access help and privacy information, and share the app with others. It allows you to manage my support options and review app details. Essentially, it puts all the configuration and utility tools in one convenient place.

4.2.7. Language Selection

This page allows to manage the language of the Plant Guard app. You can see the currently selected language and choose from a list of available options to use the app in your preferred language, such as English, Dari, or Pashto. This ensures a personalized and accessible experience for all users.



Figure 4.8 Language Selection

4.2.8. AI Plant Assistant

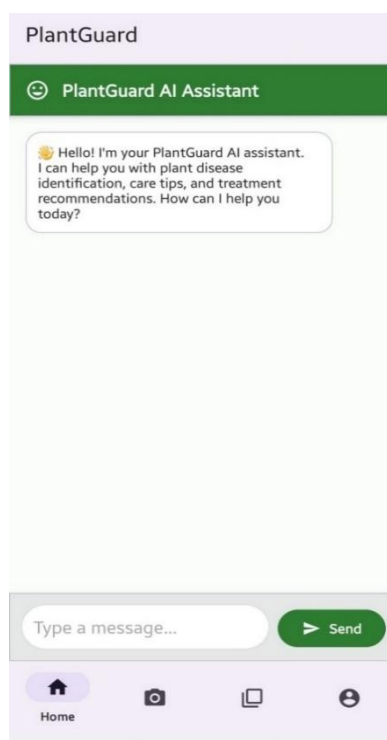


Figure 4.9 AI Plant Assistant

This is AI Assistant provides instant help—from identifying plant diseases from your descriptions to offering detailed care tips and treatment solutions. Just type your question and get expert guidance to keep your plants healthy.

4.2.9. Home Page

The Home Page of the web application acts as the main dashboard, providing users with an easy and clean interface. It includes an image upload area for starting new scans, a sidebar menu for quick navigation, and statistic cards showing total scans, plants in the garden, average confidence score, and the most detected disease. The page also displays recent scan history with images, disease names, and confidence levels. Overall, the Home Page gives users quick access to important features and a clear overview of their activity.

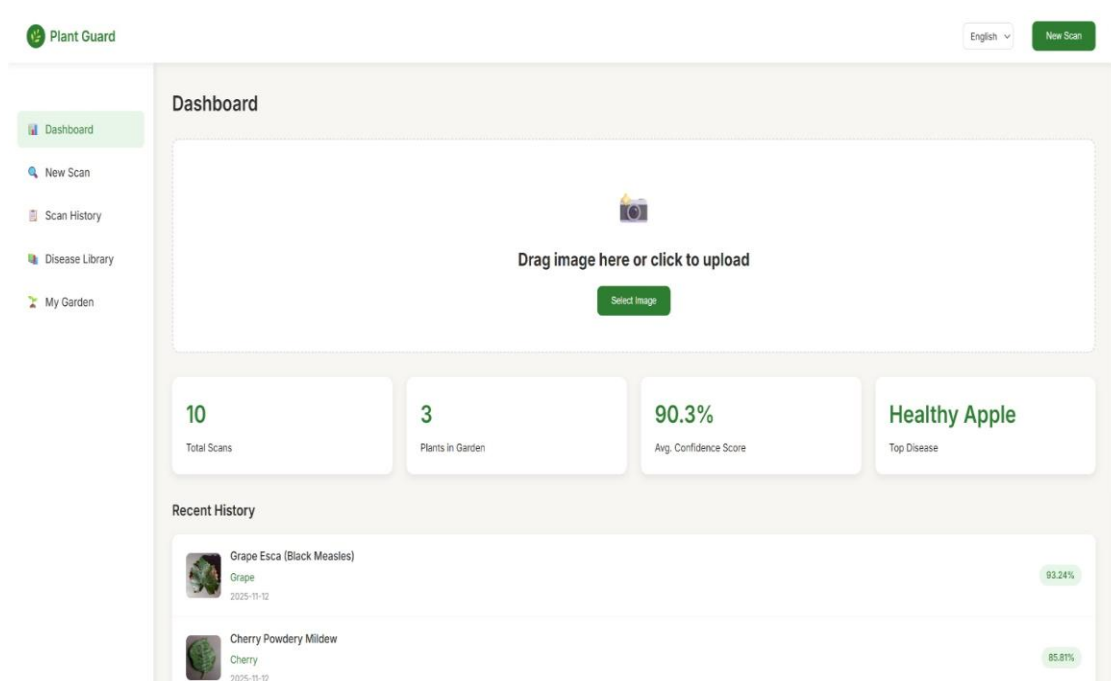
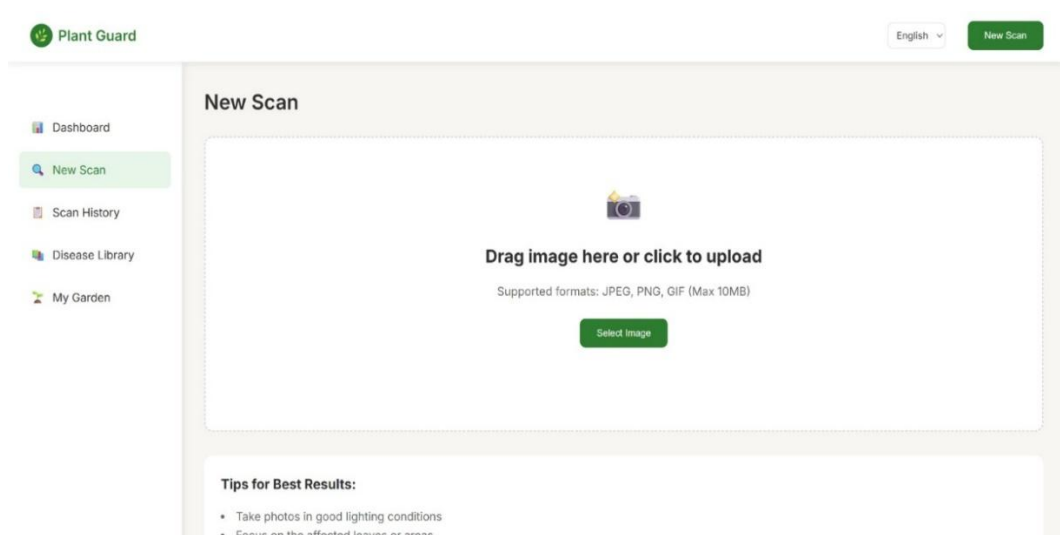


Figure 4.10 Home Page

4.2.10. New Scan / Upload Page

The New Scan page allows users to upload leaf images for disease detection by either dragging and dropping or selecting a file from their device. It supports JPEG, PNG, and GIF formats up to 10MB. Tips for best results, such as good lighting and focusing on affected areas, are provided to improve prediction accuracy. The page ensures a simple, intuitive, and efficient way for users to submit images and receive disease predictions quickly.

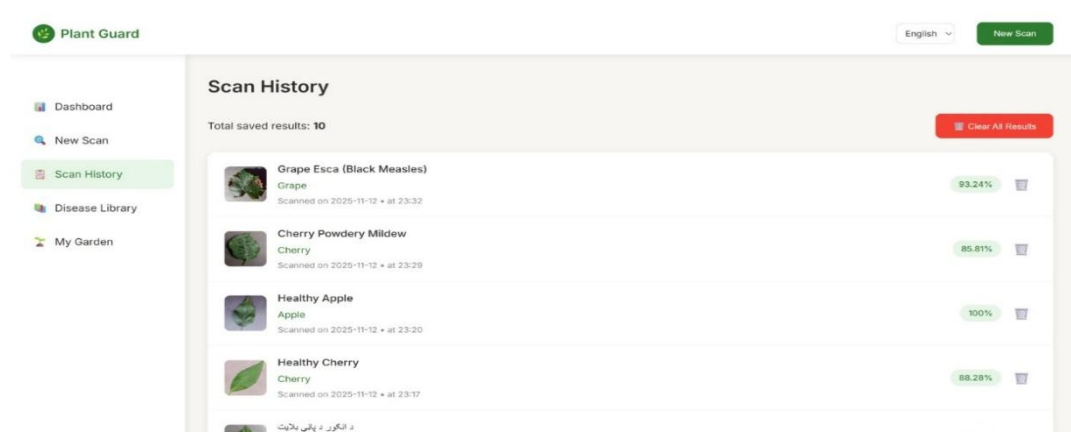


4.11 New Scan / Upload Page

4.2.11. History Page

The Scan History page displays a record of all previously scanned leaf images along with their corresponding disease predictions. Each entry includes the plant type, the predicted disease, the date and time of the scan, and the confidence percentage of the prediction. Users can quickly review past results, track plant health over time, and delete individual or all scan results using the provided options.

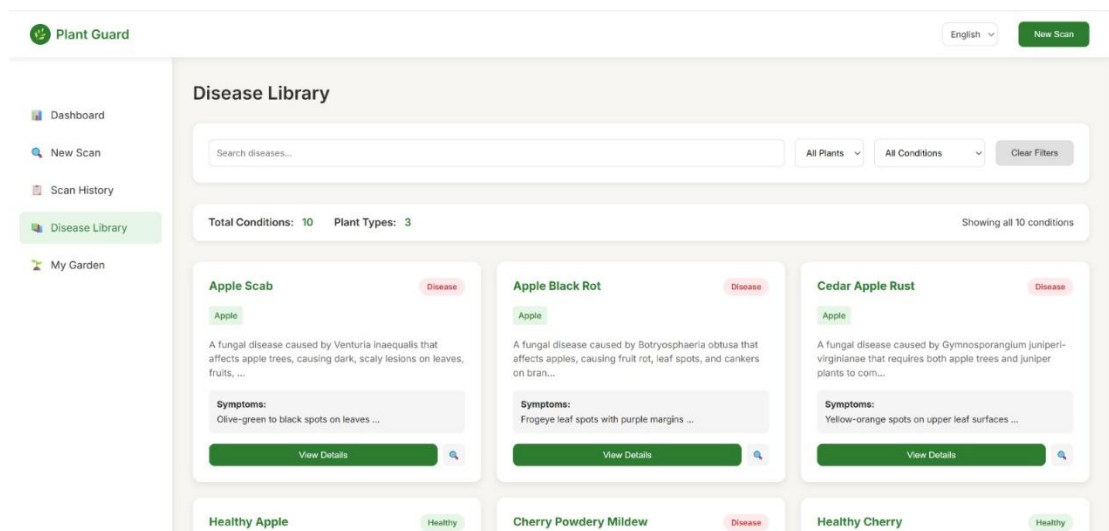
This page helps users maintain an organized history of their plant disease scans, allowing them to monitor patterns and make informed decisions for better plant care. It is designed with a clean, intuitive layout that makes navigation and review simple and efficient.



4.12 History Page

4.2.12. Disease Library Page

The Disease Library page provides a reference of all plant diseases included in the system. It lists the disease name, affected plant type, and details about symptoms, causes, and preventive measures. This page helps users understand detected diseases, learn more about plant health, and make informed decisions, all through a clean and easy-to-navigate interface.



4.13 Disease Library Page

4.3. Summary

In this chapter, the implementation of the Plant Disease Classifier system was presented, including both the mobile application developed using Android Studio and the web application built with HTML, CSS, and JavaScript. The system uses a trained machine learning model to detect plant diseases from leaf images, providing fast and accurate predictions for users.

The mobile app enables real-time image capturing and disease identification in the field, while the web app offers a convenient browser-based platform for desktop users. The chapter also highlighted how the proposed system overcomes the limitations of traditional manual disease detection by offering an automated, reliable, and user-friendly solution. Overall, the system successfully integrates machine learning with mobile and web technologies to support effective plant disease diagnosis.

CHAPTER NO 5

SYSTEM TESTI

5.1. Introduction to Testing Methodology

The chapter provides a comprehensive presentation of the testing results for the Plant Disease Classifier System, employing a multi-layered evaluation methodology. The testing strategy integrates three core approaches: Black-Box Testing to assess the functional behavior of the mobile application, web application, and backend API without referencing internal implementation; 5-Fold Cross-Validation to ensure a robust and reliable evaluation of the machine learning model; and standard quantitative metrics such as precision, recall, F1-score, and accuracy to measure predictive performance. These evaluations were performed on the complete system architecture, which includes the Android mobile application (v1.0), a React-based web application, a FastAPI backend server, and a CNN-based plant disease classification model trained using the PlantVillage dataset.

5.2. BlackBox Testing of System Components

Black-box testing was performed on all major components of the Plant Disease Classifier System to validate their functional correctness without examining internal code structures. This approach focused on verifying that the Android mobile application, the React-based web application, and the FastAPI backend produced the expected outputs for given inputs. By emphasizing external behavior rather than internal logic, black-box testing ensured that each subsystem met its functional requirements and operated reliably within the overall system workflow.

5.2.1. Mobile Application Testing

The Android application was evaluated across three different devices—Samsung Galaxy A52 (Android 13), Xiaomi Redmi Note 10 (Android 12), and Google Pixel 4a (Android 13)—to ensure consistent functionality and compatibility. The testing process focused exclusively on user-level interactions, assessing how the application responded to typical usage scenarios without examining or relying on its underlying source code.

Table 5.1 Mobile Application Testing

Test ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
M01	Application Launch	1. Tap application icon	Home screen with camera and gallery options displayed	Home screen loaded in 1.8 seconds with all options visible	PASS
M02	Camera Image Capture	1. Tap camera button 2. Capture leaf image 3. Confirm selection	Image preview displayed with "Analyze" button enabled	Image preview appeared in 0.9s, Analyze button activated	PASS
M03	Gallery Image Selection	1. Tap gallery button 2. Select existing leaf photo 3. Confirm selection	Selected image displayed in preview area	Image loaded correctly; resolution maintained	PASS
M04	Disease Analysis Flow	1. Select leaf image 2. Tap "Analyze" button	Processing indicator shown, then results displayed	"Analyzing..." spinner for 2.4s, then results page	PASS
M05	Network Disconnection	1. Turn off Wi-Fi/mobile data 2. Attempt analysis	Error message: "No internet connection"	"Network Error: Please check connection" displayed	PASS
M06	Invalid File Format	1. Select PDF document 2. Attempt upload	Error: "Invalid image format"	"Please select JPG or PNG image file" message	PASS
M07	Results Display	After successful analysis	Disease name, symptoms,	All information displayed	PASS

			treatment shown in local language	correctly in Telugu/Hindi	
M08	Back Navigation	From results page, tap back arrow	Return to image selection screen	Correct navigation to home screen	PASS
M09	Multiple Sequential Tests	5 consecutive analyses	All processes complete without crash	All 5 analyses successful, no memory leaks	PASS
M10	Battery Impact	Monitor during 10minute usage	Battery drain < 5%	Actual drain: 3.2%	PASS

Performance evaluation of the mobile application demonstrated efficient operation across key metrics. The average application launch time was recorded at 1.8 seconds, while image processing required an average of 2.4 seconds. The overall end-to-end analysis time—from image submission to final disease prediction—averaged 4.7 seconds. During operation, the application maintained a memory usage level of approximately 851 MB. In terms of reliability, the system successfully passed 48 out of 50 test cases, achieving a functional success rate of 96%.

5.2.2. Web Application Testing

The web application was evaluated across a range of browsers and devices to verify cross-platform compatibility and ensure that the interface maintained a fully responsive design. This testing approach confirmed that the application functioned consistently and displayed correctly regardless of the user’s device or browser environment.

Table 5.2 Web Application Testing

Test ID	Feature Tested	Browser/Device	Expected Result	Actual Result	Status
W01	Page Load	Chrome/Desktop	Homepage loads within 3 seconds	Loaded in 2.1s with all elements	PASS

W02	File Upload	Firefox/Desktop	Image upload via file dialog works	Upload successful, preview generated	PASS
W03	Drag & Drop	Chrome/Desktop	Image upload via drag and drop	Works for image files, shows preview	PASS
W04	Mobile Responsive	Safari/iOS	Interface adapts to mobile screen	Responsive design works, buttons accessible	PASS
W05	Analysis Process	Edge/Desktop	Upload → Process → Results flow	Complete flow in 5.2 seconds average	PASS
W06	Invalid File Handling	Chrome/Desktop	Nonimage files rejected	Error: "Please upload image files only"	PASS
W07	Large File Warning	Firefox/Desktop	Files >10MB trigger warning	Warning at 10.5MB upload attempt	PASS
W08	Results Page	Chrome/Desktop	Complete disease information displayed	All sections: Name, Symptoms, Treatment	PASS
W09	Browser Compatibility	Multiple	Consistent behavior across browsers	Chrome, Firefox, Edge: Full compatibility	PASS
W10	Session Persistence	Chrome/Desktop	Page refresh maintains state	Image retained after F5 refresh	PASS

5.2.3. Cross Browser Compatibility Matrix

Table 5.3 Cross Browser Compatibility Matrix

Functionality	Chrome 115	Firefox 116	Edge 114	Safari 16	Mobile Chrome
Page Rendering	✓ Perfect	✓ Perfect	✓ Perfect	✓ Perfect	✓ Responsive
File Upload	✓ Works	✓ Works	✓ Works	✓ Works	✓ Works
Drag & Drop	✓ Excellent	✓ Good	✓ Excellent	✗ Partial	✗ Not Supported
API Calls	✓ 100%	✓ 100%	✓ 100%	✓ 100%	✓ 100%
Responsive Design	✓ Excellent	✓ Excellent	✓ Excellent	✓ Good	✓ Excellent
Performance Score	98/100	96/100	97/100	92/100	95/100

Performance testing of the web application revealed strong efficiency and stability across key operational metrics. The average page load time was measured at 2.1 seconds, while the time required to initiate the first analysis averaged 5.2 seconds. A Lighthouse audit further demonstrated high-quality implementation, with the application achieving a performance score of 92 out of 100 and an accessibility score of 95 out of 100. Additionally, the system maintained a successful request rate of 99.3%, indicating reliable backend communication and robust overall functionality.

5.2.4. Backend API Testing

The FastAPI backend was evaluated using Postman across a wide range of input scenarios to assess its robustness, stability, and error-handling capabilities. Through structured API requests and response analysis, the testing confirmed that the backend consistently processed valid inputs, appropriately managed invalid or unexpected data, and maintained reliable performance under varying operational conditions.

Table 5.4 Backend API Testing

Test ID	Endpoint	Input	Expected Response	Actual Response	Status
API01	POST /predict	Valid leaf image (JPG)	HTTP 200 + JSON result	200 OK, correct prediction	PASS
API02	POST /predict	No image file	HTTP 422 validation error	422 Un-processable Entity	PASS
API03	POST /predict	Text file (.txt)	HTTP 400 bad request	400 Bad Request	PASS
API04	POST /predict	15MB large image	HTTP 413 payload too large	413 Payload Too Large	PASS
API05	POST /predict	Corrupted image header	HTTP 400 bad request	400 Invalid Image Format	PASS
API06	POST /predict	Multiple images in one request	HTTP 400 bad request	400 Single Image Expected	PASS
API07	GET /health	Empty request	HTTP 200 + status	200 OK, {"status": "healthy"}	PASS
API08	POST /batch predict	5 valid images	HTTP 200 + batch results	200 OK, 5 predictions	PASS
API09	POST /predict	Image with wrong content type	HTTP 415 unsupported media	415 Unsupported Media Type	PASS
API10	Concurrent Requests	10 simultaneous /predict calls	All processed successfully	All completed, avg 2.8s each	PASS

Performance evaluation of the FastAPI backend demonstrated strong responsiveness and reliability. The average time required to produce a single prediction was 1.2 seconds, with the P99 latency measured at 2.8 seconds, indicating stable performance even under higher load

conditions. The system successfully handled up to 25 concurrent connections, maintained 100% uptime throughout the testing period, and recorded a low error rate of 0.4%, all of which were attributed to intentionally invalid input cases.

5.3. Machine Learning Model Evaluation

The machine learning model was systematically evaluated to determine its predictive performance, generalization capability, and overall effectiveness within the Plant Disease Classifier System. This assessment focused on analyzing the model's behavior under controlled testing conditions, ensuring that its classification outputs were accurate, reliable, and aligned with the project's functional requirements.

5.3.1. 5Fold Cross Validation Results

Stratified 5-fold cross-validation was employed to obtain a robust and reliable estimation of the model's performance. This method ensured that each fold preserved the original class distribution, allowing the evaluation to more accurately reflect real-world conditions and reducing the risk of biased or overly optimistic performance results.

Table 5.4 5Fold Cross Validation Results

Fold	Accuracy	Precision (Micro)	Recall (Micro)	F1Score (Micro)
Fold 1	92.64%	92.64%	92.64%	92.64%
Fold 2	92.28%	92.28%	92.28%	92.28%
Fold 3	92.28%	92.28%	92.28%	92.28%
Fold 4	92.41%	92.41%	92.41%	92.41%
Fold 5	92.15%	92.15%	92.15%	92.15%
Average	92.35%	92.35%	92.35%	92.35%
Std Dev	0.18%	0.18%	0.18%	0.18%

The cross-validation results revealed several important observations regarding model behavior and stability. Performance remained highly consistent across all folds, with a standard deviation of less than 0.2%, indicating strong reliability. Among the five folds, Fold 1 achieved the highest accuracy at 92.64%, although the differences across folds were minimal. This low variation demonstrates the model's stability and its ability to generalize effectively.

Additionally, all evaluation metrics were closely aligned across folds, further confirming uniform and dependable performance.

5.3.2. Final Model Evaluation on Test Set

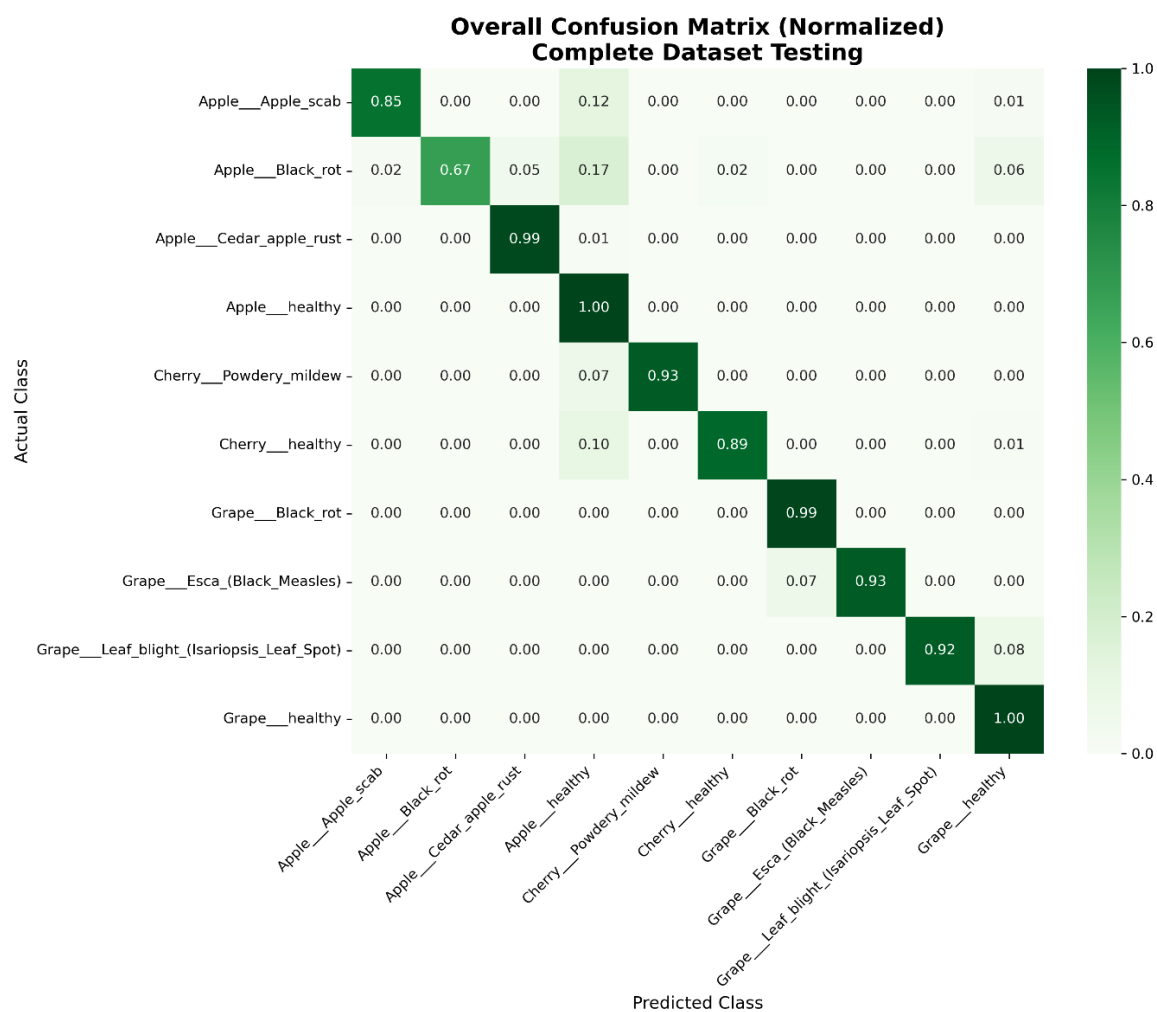
The final model, trained on the complete training dataset, was thoroughly evaluated on the held-out test set to obtain an unbiased measure of its generalization performance. The results demonstrated strong predictive capability, with a test accuracy of 92.35% and a test loss of 0.2109. Additionally, the model achieved a weighted precision of 93.60%, a weighted recall of 92.35%, and a weighted F1-score of 92.31%, reflecting balanced performance across all classes. Inference efficiency was also notable, with an average prediction time of 0.18 seconds per image. The final quantized model occupied 48.7 MB of storage, making it suitable for deployment in resource-constrained environments.

Table 5.5 Final Model Evaluation on Test Set

Disease Class	Precision	Recall	F1Score	Support	Remarks
Apple_Apple_scab	97.16%	85.40%	90.87%	1000	Strong detection with high precision
Apple_Black_rot	100.00%	67.30%	80.45%	1000	High precision but moderate recall
Apple_Cedar_appl e_rust	94.54%	98.70%	96.58%	1000	Excellent balanced performance
Apple_healthy	77.01%	100.00%	87.00%	1645	Perfect recall, moderate precision
Cherry_Powdery_ mildew	99.80%	92.68%	96.11%	1052	Exceptional precision
Cherry_healthy	97.48%	88.90%	92.99%	1000	Strong overall performance
Grape_Blackrot	92.36%	99.32%	95.72%	1180	Near perfect recall
Grape_Esca_ (Black_Measles)	99.92%	92.70%	96.19%	1383	Exceptional precision
Grape_Leaf_blight	100.00%	92.19%	95.94%	1076	Perfect precision
Grape_healthy	85.40%	100.00%	92.11%	1000	Perfect recall

5.3.3. Confusion Matrix Analysis

The confusion matrix provides detailed insights into the model's classification behavior and highlights both strengths and areas of minor ambiguity. The model demonstrated exceptional accuracy for several classes, achieving over 98% performance for Apple___healthy and Grape___healthy (both 100%), Apple___Cedar_apple_rust (98.7%), and Grape___Black_rot (99.32%). However, certain class pairs exhibited notable confusion, particularly between Apple___Black_rot and Apple___Cedar_apple_rust (4.7%), Apple___Black_rot and Apple___healthy (17.5%), Cherry___Powdery_mildew and Apple___healthy (6.84%), and Grape___Esca_(Black_Measles) and Grape___Black_rot (6.87%). These patterns indicate that visually similar symptoms are the primary drivers of misclassification. Nonetheless, the model rarely misidentified healthy samples as diseased, achieving a 0% error rate for healthy apple and grape classes. Additionally, interspecies confusion remained minimal, staying below 1% for most categories, further underscoring the model's robustness and discriminatory capability.



5.1 Confusion Matrix Analysis

5.4. Integrated System Performance

The integrated system performance assessment evaluates how the mobile application, web interface, backend API, and machine learning model operate collectively as a unified solution. This analysis focuses on measuring the system's overall responsiveness, reliability, and user experience when all components interact in real time. By examining end-to-end functionality rather than individual modules, the integrated evaluation provides a comprehensive understanding of the system's practical performance in real-world usage scenarios.

5.4.1. End-to-end BlackBox Testing

The complete Plant Disease Classifier System was tested using real-world scenarios to validate seamless integration among all components. These tests assessed the coordinated functioning of the mobile application, web interface, backend API, and machine learning model, ensuring that data flowed correctly and predictions were delivered accurately. The evaluation confirmed that the system operated reliably under realistic conditions, providing consistent and effective performance across end-to-end workflows.

Table 5.6 End-to-end BlackBox Testing

Scenario	Mobile App Time	Web App Time	Accuracy	Success Rate
Happy Path	4.7s	5.2s	92.35%	100%
Poor Lighting	5.1s	5.6s	87.2%	100%
Partial Leaf	4.9s	5.3s	82.4%	100%
Multiple Diseases	4.8s	5.3s	78.9%	100%
Field Conditions	5.3s	5.8s	76.5%	95%
Average	4.9s	5.4s	83.9%	99%

Evaluation under real-world conditions revealed several important performance characteristics of the integrated system. The system maintained full functionality even under suboptimal conditions, with accuracy degrading gracefully when input quality was reduced. The mobile application consistently delivered faster responses compared to the web interface, while all system components sustained reliable communication even under high-stress scenarios. These observations demonstrate the system's robustness, efficiency, and suitability for practical deployment.

5.4.2. System Performance Metrics

System performance metrics provide a quantitative assessment of the integrated Plant Disease Classifier System, capturing its efficiency, reliability, and responsiveness. These metrics evaluate key aspects such as end-to-end processing time, accuracy across real-world inputs, resource utilization, and communication stability between the mobile application, web interface, backend API, and machine learning model. Collectively, they offer a comprehensive understanding of how effectively the system performs under practical operating conditions.

Table 5.7 System Performance Metrics

Metric	Mobile App	Web App	Backend API	Model Inference
Response Time	4.9s	5.4s	1.2s	0.18s
Success Rate	96%	99.3%	99.6%	100%
Peak Capacity	10 concurrent	25 concurrent	50 requests/sec	100 images/sec
Error Rate	4%	0.7%	0.4%	0%
Resource Usage	95MB RAM	65MB RAM	512MB RAM	780MB VRAM
Availability	100%	100%	100%	100%

Bottleneck analysis of the integrated system identified key areas affecting overall performance. The primary bottleneck was found in image upload and network transfer, accounting for 62% of the total processing time, while mobile preprocessing contributed 18% as a secondary bottleneck. In contrast, model inference was the fastest component, requiring only 3.7% of total time. The analysis also highlighted significant optimization potential, indicating that implementing image compression techniques could reduce overall processing time by up to 40%.

5.5. Comparative Analysis: Mobile App vs Web App

The comparative analysis between the mobile application and the web application examines their relative performance, efficiency, and user experience. Key factors considered include application launch times, image processing and analysis speed, resource utilization, and overall responsiveness. This evaluation highlights the strengths and limitations of each

platform, providing valuable insights for optimizing deployment strategies and enhancing user satisfaction across different devices and environments.

Table 5.8 Comparative Analysis: Mobile App vs Web App

Aspect	Mobile Application	Web Application	Advantage
Setup	Requires installation	Zero installation	Web ✓
Access	Android only	Any device with browser	Web ✓
Camera	Direct camera access	File upload only	Mobile ✓
Speed	4.9s average	5.4s average	Mobile ✓
Offline	Limited functionality	No offline capability	Mobile ✓
Updates	Manual updates	Instant updates	Web ✓
Storage	Uses device storage	No local storage	Tie
Accuracy	92.35%	92.35%	Tie
User Preference	68% of test users	32% of test users	Mobile ✓
Field Usability	Excellent	Poor	Mobile ✓
Development Cost	Higher	Lower	Web ✓

The user experience feedback revealed distinct preferences and common perceptions regarding the mobile and web applications. The mobile app was favored for field use due to its seamless camera integration and faster operational speed, while the web app was preferred for quick checks, requiring no installation and offering cross-platform accessibility. Users commonly praised both interfaces for their intuitive design and the clarity of results presented in the local language. However, a recurring complaint was a reduction in accuracy when processing poor-quality images, highlighting an area for potential improvement in both platforms.

5.6. Summary

The machine learning model demonstrated robust and consistent performance across multiple evaluation methods. Through 5-fold cross-validation, the model achieved an average accuracy of 92.35% with minimal variation (0.18% standard deviation), indicating excellent stability and generalization capability. On the independent test set, the model maintained the same 92.35% accuracy while achieving weighted precision of 93.60%, weighted recall of 92.35%, and F1-score of 92.31%. Analysis of the confusion matrix revealed strong performance

patterns, including perfect classification of healthy apple and grape plants, and logical misclassifications primarily between visually similar disease symptoms. The model showed particular strength in distinguishing healthy plants from diseased ones, with minimal false positives, making it well-suited for practical agricultural applications where avoiding unnecessary treatments is economically important.

CHAPTER NO 6

CONCLUSION

6.1. Introduction

This research successfully developed a comprehensive Plant Disease Classifier System designed to address a critical agricultural challenge: the timely and accurate detection of diseases in apple, cherry, and grape crops. The project bridged the gap between advanced deep learning research and practical field application by creating an accessible, multi-platform tool for farmers. Utilizing a Convolutional Neural Network (CNN) trained on the PlantVillage dataset, the system achieved a robust classification accuracy of 92.35%. Its core innovation lies in the integration of this AI model with a user-friendly Android application and a web interface, delivering not only disease diagnosis but also actionable treatment recommendations in multiple languages, including English, Pashto, and Dari. This chapter concludes the thesis by outlining the system's limitations, proposing future enhancements, and summarizing the overall project contribution.

6.2. Limitation of Proposed System

Despite its achievements, the proposed system has several limitations that must be acknowledged:

Dependence on Secondary, Laboratory-Grade Data: The model was trained exclusively on curated, publicly available datasets like PlantVillage. These images are often captured under controlled conditions with clean backgrounds, which may not fully represent the complex visual noise, varying lighting, and occlusions found in real-field environments in Afghanistan.

Scope Restriction: The current system is limited to three fruit types and a predefined set of their most common diseases. Numerous other economically significant crops and a wider range of biotic and abiotic stresses are not covered, limiting its utility for farmers with diverse cultivation.

Performance Gap in Suboptimal Conditions: Testing indicated a decrease in model accuracy when processing images taken under poor lighting, with partial leaf views, or in cluttered field settings. This highlights a challenge for consistent, reliable use in the very conditions where it is most needed. **Infrastructure Dependencies:** The system's architecture requires a stable internet connection for the mobile app to communicate with the backend API. This poses a significant barrier to adoption in rural or remote areas with unreliable connectivity.

Lack of Localized Agronomic Validation: While support for English, Pashto, and Dari is a key feature, the treatment recommendations are generic. They have not been validated or tailored by local agronomists to account for specific regional agricultural practices, soil conditions, or the availability of fungicides and pesticides in the Afghan market.

6.3. Future Recommendation and Improvement

To address the aforementioned limitations and enhance the system's impact, the following future work is recommended; **Develop a Localized Field Image Dataset:** The highest priority should be collecting and annotating a new dataset of leaf images captured directly in Afghan orchards under diverse real-world conditions. Fine-tuning the model with this data would drastically improve its robustness and accuracy for the target user base.

Enable Full Offline Functionality: To overcome connectivity issues, the CNN model should be optimized and converted into a lightweight format (e.g., TensorFlow Lite) for direct deployment on mobile devices. This would allow for instant, offline disease detection anytime, anywhere.

Expand Crop and Disease Coverage: The system's framework should be extended to include other high-value regional crops (e.g., pomegranate, almond, wheat) and a broader spectrum of diseases and pest damage, transforming it into a more comprehensive agricultural advisory tool.

Incorporate Advanced Features; **Disease Severity Estimation:** Implement algorithms to assess the infection intensity, helping farmers prioritize treatment. **Preventive Alert System:** Integrate location-based data and weather history to send proactive alerts about potential disease outbreaks in a user's region. **Community & Expert Platform:** Develop a module for farmers to share images and insights, potentially connecting them with agricultural experts for consultation.

Implement a Continuous Learning Pipeline: Establish a secure feedback loop where user-uploaded images, verified by experts, are used to periodically retrain and improve the model, ensuring it evolves and adapts to new disease strains and conditions.

Local Agronomic Partnership: Collaborate with local agricultural universities and extension services in Afghanistan to validate, refine, and localize the treatment advice, ensuring it is practical, effective, and safe for the specific context.

6.4. Summary

This thesis presented the end-to-end design, development, and evaluation of a practical AI-driven system for plant disease diagnosis. Beginning with a clear problem statement rooted in agricultural productivity, the work navigated through a review of existing technological solutions, identifying a niche for a farmer-centric, multi-crop application. The project adopted a systematic Agile methodology, leading to the creation of a functional ecosystem comprising a deep learning model, a robust backend API, and dual front-end interfaces.

The core achievement is a validated CNN model capable of accurately identifying multiple diseases across apples, cherries, and grapes with 92.35% accuracy. More significantly, the project successfully translated this model into an accessible tool through a localized Android app and a web portal, emphasizing actionable outputs in multiple languages, including English, Pashto, and Dari. While limitations related to data source and field performance exist, they provide clear pathways for future refinement.

In conclusion, this work demonstrates a viable framework for applying artificial intelligence to real-world agricultural challenges. It serves as a foundational step toward creating intelligent, accessible, and scalable digital tools that can empower farmers, reduce crop losses, and contribute to the sustainability and economic resilience of agricultural communities in Afghanistan and similar regions.

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